Paul (Yi-Chia) Chen
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EDUCATION

University of Southern California, Los Angeles, CA — Aug 2024 – Aug 2028
Ph.D. in Electrical and Computer Engineering – advised by Prof. Christopher Torng
Research Focus: Compiler-Directed Optimization for Energy-Efficient Machine Learning Accelerators and Silicon Prototyping

University of Southern California, Los Angeles, CA — Aug 2023 – May 2024
M.S. in Computer Engineering

University of Southern California, Los Angeles, CA — Aug 2019 – May 2023
B.S. in Electrical and Computer Engineering (W.V.T. Rusch Honors Program Research Award)

PUBLICATIONS

ICCAD 2026 J. Kim, S. Yarzada, **P. YC. Chen**, Y. Li, W. Zhu, and C. Torng. "Chipstitch: A Scalable and Cost-Efficient Platform for Accelerated Silicon Prototyping." IEEE/ACM International Conference on Computer-Aided Design (ICCAD). November 2026.

ISLPED 2026 **P. YC. Chen**, J. Kim, W. Zhu, Y. Li, S. Huang, C. Weng, and C. Torng. "PowerFlow-DNN: Compiler-Directed Fine-Grained Power Orchestration for End-to-End Edge AI Inference." IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED). August 2026. ***Best Paper Candidate.***

FPL 2023 **P. Chen (*)**, P. Manjunath (*), S. Wijeratne, B. Zhang, and V. Prasanna. "Exploiting On-Chip Heterogeneity of Versal Architecture for GNN Inference Acceleration." IEEE International Conference on Field-Programmable Logic and Applications (FPL). September 2023. (* = denotes equal contribution)

SKILLS

- **PDK and Flow Integration:** TSMC 40 nm LP, TSMC 28 nm HPC+, Intel 16 nm
- **Hardware Design:** Verilog/SystemVerilog, RTL design/verification, CDC, SDF GLS, low-power ASIC design, FPGA
- **ASIC Implementation**: Logic synthesis, physical design, STA, CTS, power planning, RTL-to-GDS
- **EDA Tools**: Cadence Genus/Innovus, Synopsys VCS/PrimeTime/Verdi, Siemens Calibre
- **Programming/ML:** Python, C++, CUDA, PyTorch
- **Architecture:** Computer architecture, ML accelerators, hardware/software co-design, CGRAs, GPGPU-Sim, gem5

SELECTED RESEARCH AND SILICON PROJECTS

Silicon Query Architecture Platform – *TSMC 28 nm HPC+, Intel 16 nm* — Aug 2025 – Present
USC ACORN Lab | ICCAD 2026 | ASPLOS 2027 submission

- Architected a polymorphic on-chip network providing shared test infrastructure, per-site bandwidth guarantees, programmable clocking, and independent power-domain control; described in a co-first-author ASPLOS 2027 submission.
- Designed and verified parameterized SystemVerilog RTL for CDCs, asynchronous FIFOs, MMIO control, and 64/128/256-bit datapath-width conversion using Synopsys VCS and Verdi.
- Developed CTS and clocking strategies for heterogeneous silicon-query block integration, improving scalability of physical integration and timing closure in both Intel 16 nm and TSMC 28 nm HPC+; published at ICCAD 2026.
- Led a TSMC 28 nm HPC+ RTL-to-GDS flow using Cadence Genus and Innovus, including floorplanning, power planning, multi-clock constraints, CTS, STA, and timing closure at up to 3 GHz; targeting a November 2026 tapeout.

PowerFlow-DNN Accelerator – *TSMC 40 nm* — Dec 2025 – Present
USC ACORN Lab | ISLPED 2026 Best Paper Candidate | TVLSI 2027 submission

- Developed a compiler-directed framework that jointly schedules DVFS and power gating across accelerator domains, reducing inference energy by up to 48% on a custom AI accelerator under application-level performance constraints.
- Implemented and verified a SystemVerilog hybrid SRAM/RRAM accelerator with multiple voltage and power-gated domains through a shuttle-targeted RTL-to-GDS flow.

- Completed synthesis, place-and-route, STA, SDF-annotated gate-level simulation, and post-layout power characterization to construct the timing and energy models used in the ISLPED 2026 evaluation with a supporting paper artifact.
- Extended PowerFlow-DNN from periodic single-network inference to multi-stage world-model planning across heterogeneous workloads; submitted to TVLSI.

Multi-Precision LLM Accelerator – *Intel 16 nm Tapeout* Aug 2024 – Dec 2024

USC Advanced Chip Design and Tapeout (EE 599)

- Designed a BERT-layer accelerator supporting configurable 4-, 8-, and 16-bit QKV datapaths for evaluating accuracy and energy tradeoffs; taped out in Intel 16 nm.
- Led a six-person team through an RTL-to-GDS flow and SDF-annotated GLS; achieved timing closure at 1 GHz.

GNN Accelerator on Xilinx Versal ACAP Jun 2022 – Jun 2023

USC FPGA/Parallel Computing Lab | FPL 2023

- Developed a hardware/software co-design for GNN inference on heterogeneous Xilinx Versal ACAP, achieving a 162× speedup over the CPU baseline; published at FPL 2023.
- Designed workload-partitioning and data-movement strategies across programmable logic and AI Engines for sparse and dense operations, reducing end-to-end inference latency.

INDUSTRY EXPERIENCE

Machine Learning Engineer Intern Jul 2023 – Jul 2024

EMD Electronics (Merck KGaA) | Carlsbad, CA

- Developed linear, XGBoost, and random-forest models to identify semiconductor process features associated with yield changes; optimized distributed data-processing and ML workflows, reduced overall compute cost and improved efficiency.
- Implemented automated retraining and model-monitoring pipelines on Palantir Foundry for evolving semiconductor-manufacturing data.

TEACHING EXPERIENCE

Teaching Assistant, Advanced Chip Design and Tapeout (EE 599) – Fall 2024, Fall 2025

- Mentored 30 students across six teams through RTL-to-GDS implementation and verification, producing six designs integrated into an Intel 16 nm tapeout.

Course Lab Mentor, Introduction to Digital Circuits (EE 354) – Spring 2022

- Guided students through FPGA-based digital-circuit labs and hosted weekly office hours covering RTL development, simulation, and hardware debugging.

Course Grader, Introduction to C++ (EE 155) – Fall 2021

- Graded student C++ assignments, labs, and exams.